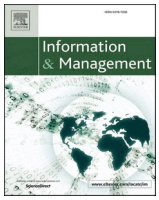
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Effect of Designer- versus User-driven Network-monitoring Dashboard Design on User Flow Experience and Performance: The Role of Augmented Virtuality

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ABSTRACT

A well-designed dashboard can effectively communicate critical information to network administrators and enable them to identify and solve network issues quickly. However, little is understood about how network-monitoring dashboard designs impact user experience in an augmented virtuality (AV) environment. Our study proposes that a dashboard design showing variations in selected elements can impact the network-monitoring flow experience of users by affecting their enjoyment, concentration, and control, which in turn can influence their performance in critical, complex, and time-sensitive environments. To investigate this phenomenon, we design and evaluate two narrative visualization user interfaces—a designer-driven (DD) design and a user-driven (UD) design—in the context of network-monitoring dashboards. Adding vivid and familiar real-world objects may enhance perceived realism; thus, we integrate AV with the two dashboard designs (i.e., DD and UD) and evaluate their effect on user flow experience and performance. Results suggest that the flow experience of users will lead to positive perceived performance. A nonlinear relationship between AV and dashboard design, and the UD design with a moderate AV level can improve user flow experience and performance compared with the DD design. When the AV level is nonexistent or excessive, the superior effect of the UD design disappears. Lastly, we discuss practical implications and theoretical contributions, along with future research directions.

1. Introduction

Despite the tremendous efforts exerted to enhance the robustness of information technology (IT) infrastructure, network failures frequently occur and can cause huge financial repercussions, as evidenced by the outages at Amazon Web Services (AWS), T-Mobile, and Facebook [1]. For example, the AWS outage in Dec 2021 impacted not only Amazon's delivery and fulfillment but also other apps and websites that rely on AWS (Vemno, Roomba vacuum, and Ring doorbells). The outage also adversely impacted Delta flights, Disney theme parks (guests were unable to check-in online), and day traders (Robinhood and Coinbase were offline). Subsidiaries such as WhatsApp, Instagram, and Oculus were also unavailable during what was arguably the largest system outage in history [1]. The number of incidents has not decreased, regardless of additional investments, resulting in short- and long-term impacts, such as damaged reputation, loss of customers, and reduced profitability. A critical IT outage is a time-sensitive and stressful work environment for IT professionals and can impact their productivity, stress level, and how they interact with information for making decisions [2].

Network managers can be equipped with a well-designed network-monitoring dashboard that provides them with network visibility into operational aspects to improve uptime. Such a network-monitoring dashboard can provide real-time data on devices and bandwidth utilization and include an alert mechanism that displays information on the health of the network. A well-designed dashboard can increase network managers' probability of identifying problems before they become disasters and allow them to troubleshoot and fix problems quickly after their occurrence. Hence, IT professionals can rely on well-designed network-monitoring dashboards to effectively perform critical, complicated, and time-sensitive problem-solving tasks in stressful

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environments [3].

A dashboard is a visualization interface that provide the operational status of all the elements monitored by a network-management system. Interface-related design choices can play a crucial role in the usability and effectiveness of information systems (IS) [4]. An effective dashboard design will allow network engineers to proactively address service outage warnings [5], whereas a dashboard that is not user-friendly can impede users from identifying or addressing network issues in a timely manner. When dashboard information is presented in an unorganized manner, it hinders the ability of the end user (e.g., network engineer, network administrator, and other IT support staff) to proactively monitor and take appropriate action to maintain the network status [6]. Visualized data can be presented in a linear format or in an interactive manner to support flexible data exploration. Although Segel and Heer [7] *conceptually* classified narrative visualizations along a spectrum of designer-driven (DD) and user-driven (UD) designs, empirical research comparing the differences between the effectiveness of DD and UD designs is limited.

The literature has yet to reach a consensus on which approach (i.e., the UD design or DD design) is superior in the context of network monitoring. Segel and Heer [7] suggested that a DD design is best suited for efficient communication, whereas a UD design is best suited for data diagnostics. Similarly, some scholars argued that a DD design can efficiently communicate information [8], because it provides complete information at one time [9], whereas others (e.g., [10]) argued that a UD design allows users to engage with the data and discover patterns. In the context of network monitoring, the efficient communication of the network device status and data diagnostic capability is important. However, which approach (i.e., UD design or DD design) is superior under what types of conditions remains unclear. Our research aims to contribute to the nascent literature in this domain.

The traditional human-centered design approach is well-established and promotes user interactivity to engage users [11–13]. However, this approach is being challenged by the latest futuristic reality-based technologies, which are equipped with visualization and data manipulation capabilities for utilitarian tasks. Unfortunately, limited guidelines are available on how reality-based interfaces can support digital work conducted in critical cyberinfrastructures, such as addressing critical and emergent network outage problems. This study examines exciting reality-related technological growth areas in the tech field, such as augmented virtuality (AV), which is projected to experience considerable growth. Although most reality-based applications are hedonic, that is, for entertainment purposes, their use in academia and business is on the rise [14], and AV-related designs are considered to be a novel open area for research and industry product development. Compared with studies on augmented reality (AR; [15–17]) and virtual reality (VR; [18–20]), those on AV are in the inception stage [21]. Accordingly, how the inclusion of AV elements can impact the interaction of users with work-based systems is little understood. Therefore, how user experience (UX) may be affected by the integration of AV with network-monitoring dashboard designs is an open and interesting line of inquiry.

In this study, we build on the work of Segel and Heer [7] and flow theory ([22]) and its application in the context of IS (e.g., [23–26]) to fill the key research gaps. We evaluate how different network-monitoring dashboards, namely, *DD* and *UD designs,* with three AV levels (none, moderate, and high), indicating user interaction with different degrees of virtual and real-world elements in a virtual space [27], can affect user flow experience and performance.

2. Literature Review

We review the literature on dashboard design types in Sections 2.1 (DD and UD designs) and 2.2 (AV). Then, we review the theoretical background in Section 2.3 (flow theory).

2.1. Literature on DD and UD Dashboard Designs

Data visualization is commonly used to communicate data and information. An emerging trend in this space is the combination of narrative and storytelling [28], that is, narrative visualization. Dashboards use a tiled chart layout to display multiple dimensions of the data to monitor interactive and real-time data changes and connections; thus, they are among the most commonly used narrative visualizations for decision analysis and real-time monitoring [29,30].

Segel and Heer [7] identified patterns and structures used by news media to introduce storytelling in visualizations. They presented several narrative visualization genres with varying levels of interactivity and messaging to generate UX. In other words, variations in narrativity and interactivity components allow for the construction of different visual structures [31]. Recent studies (e.g., [32]) identified the lack of narrative elements as one of the key reasons for the ineffectiveness or failure of many DD dashboards. Specifically, Fernandez-Nieto et al. [32] proposed that narrative attributes can add value, such as the arrangement of plot or content on a dashboard design with narrative structure tactics to enable effective data communication and explorative analysis. Although the concept of the narrative structure originates from the news and media domain, the inclusion of narrative attributes in a dashboard may help provide data insights, increase user engagement, and enhance the use of content guidance and thus make visualization highly expressive in the context of learning analytics dashboard design [32]. However, the inclusion of relevant and useful narratives in the dashboard design may be challenging, because designs are domain-specific. In the network-monitoring domain, narrative structures in the dashboard design are critical for engineers to understand the connected content, comprising characters, actions, and settings, which are desirable for effective human communication and interaction, as well as data semantics and sentiment encoding [33]. In an urgent situation resulting from a network outage incident, effective narrative structures in a dashboard design are crucial for supporting network engineers' problem-solving process. Therefore, exploring and evaluating narrative structure types for dashboard designs is imperative to fill the research gap in this domain.

However, research on evaluating which dashboard design narrative structure type is superior is scarce. In a recent review of the literature on dashboards over the period of 2017–2021, Revina [34] identified 31 studies that included an empirical assessment of dashboard design elements. Most of the studies focused on dashboards with no variations in the design elements, that is, the design elements have a single state (i.e., basic interactivity). In addition, such studies focused on one of the two key elements, namely, interactivity or data semantics (e.g., alerts for anomalies and automatic updates). The review revealed the limited variation in the level of interactivity of the dashboards in most of the studies and identified a similar trend in data semantics. The review also noted that prior studies examined such elements separately, instead of exploring how variations in both elements in a single design can influence UX. High interactivity and data semantics can offer users an engaging experience, which may result in improved usage. Thus, the literature review concluded that UX improvement through the modification of dashboard characteristics must be thoroughly examined and requires considerable attention from the research community and practitioners [34].

Narrative structure tactics are one of the key design dimensions of data visualization, which are related to (i) ordering, (ii) messaging, and (iii) interactivity mechanisms [7]. According to Segel and Heer [7], *ordering* represents the arrangement of the path available to the end user on a visualization dashboard, which can be linear, random, or user-directed. *Messaging* refers to the method through which observations and comments are communicated to the end user, and common messaging mechanisms are message length, labels, and captions. *Interactivity* represents the multiple methods through which the user can manipulate the visualization dashboard, and common interactivity

design elements are filtering, navigation buttons, selection, and drill-down [7]. The difference between ordering and interactivity is that ordering involves the overall arrangement of the visualization dashboard, whereas interactivity involves the manipulation of specific dashboard components, such as filtering and navigation buttons. Thus, different dashboard designs can be developed by manipulating the level of the design elements, such as ordering, messaging, and interactivity. Such design elements are aligned with the interactivity and data schematics highlighted by studies that examined the design aspects of other IS types, such as websites [35] and decision support systems [36].

Segel and Heer [7] also suggested that the three design elements (ordering, messaging, and interactivity) can be manipulated to develop two contrasting visualization designs, namely, (i) author-driven and (ii) reader-driven designs. In practice, such designs are referred to as UD (user-driven; referred to as reader-driven by Segel and Heer) and DD (designer-driven; referred to as author-driven by Segel and Heer) [31]. Therefore, in this study, we refer to the two designs: (i) UD design and (ii) DD design. A DD design follows a linear path throughout the visualization space, with minimal interactivity, and relies extensively on messaging. Examples of DD designs include films, noninteractive slideshows, and educational videos. In such cases, the audience cannot change the presentation order and will not have an option to interact with the presentation materials. However, the audience can rely on messages to understand the visualization. With regard to their narrative structure, DD dashboards are well suited to business contexts, wherein the top-level or overall view of the network status must be conveyed to the end user [7,37]. By contrast, a UD design has no predefined ordering and has limited messaging content but offers a high degree of interactivity. Examples of UD designs include visual analysis tools, such as Tableau. With a UD design, users are not constrained by the predefined order of the visualization and are free to interactively explore the data. Users may also rely less on messages to grasp the essence of the visualization. Such an approach is well suited for business tasks that focus on pattern identification, drill down, followed by an in-depth examination of the data, and hypothesis formation. The dashboard in Appendix A presents elements associated with the UD and DD approaches.

DD and UD designs have distinctive ways of engaging users in their problem-solving. However, which design is superior in the context of network-monitoring dashboard design remains unclear. Thus, we intend to compare the two visualization approaches (Table 1). The main features of the DD design are (i) linear ordering, (ii) a high degree of messaging, and (iii) a low degree of interactivity. Meanwhile, the main features of the UD design are (i) a low degree of ordering, (ii) a low degree of messaging, and (iii) a high degree of interactivity.

2.2. Literature on AV design

Milgram and Kishino [38] proposed a reality–virtuality continuum (Fig. 1), ranging from completely real environments on one side to entirely virtual environments on the opposite side. VR is typically positioned at the right end of the continuum, representing an environment where users are completely immersed in and interact with a computer-generated virtual world as if the environment was real [39]. VR typically requires the use of head-mounted displays, motion controllers, and other hardware to simulate a fully immersive environment [39]. AR and AV are located between the real world and computer graphics, blended into reality in real-time [40], with AR being closer to the real world, and AV being closer to the virtual world [21]. According to Spacey [27], the differences between AR and AV lie in the interaction of physical reality and VR. AR technology adds virtual elements to reality to enhance the real world with a virtual overlay [15]. Conversely, AV involves the addition of real-world elements to a virtual environment to enhance users' virtual experience [41] and increase their sense of presence, especially in terms of perceived visual realism [21]. AR and AV incorporate representations of virtual and real-world elements differently, with AR leaning toward reality and AV leaning toward virtuality (Fig. 1). Although both technologies represent the current state of real-world environments, the difference between the two boils down to *where user interaction occurs*. If user interaction occurs in the real world, then AR is involved, such as the Pokémon Go game [42]. However, if user interaction occurs in a virtual space, then AV is involved, such as visualization of a model of an engine in flight (the model is on the computer screen, but the real-world elements are physically thousands of kilometers away).

Table 1
Design Levels of Dashboard

	Ordering	Messaging	Interactivity
DD design	High	High	Low
UD design	Low	Low	High

AV systems can generate a view for the end user, combining elements that can be seen in the real world with virtual elements generated by a computer [21,43]. Such a combination will result in enhanced realism factors, including scene realism, consistency with the objective world, and experience meaningfulness, which can lead to increased presence. Perceived scene realism does not necessarily refer to real-world objects but to the extent to which "the information conveyed by a virtual environment is [consistent] with that learned from real-world experience" ([44], p. 230). The addition of vivid and familiar items from the real world to a virtual world in AV can increase users' perceived realism, thereby enhancing their sense of presence [21], that is, an individual's internal state of "being there" [45]. An AV environment augmented with vivid and familiar objects can make the UX of interacting with intelligent virtual agents highly immersive and thus reduce stress compared with a VR environment [21]. Therefore, AV can induce a highly optimal flow experience. The main goal of AV is to provide a visualization to minimize the gap between the virtual and real environments. In our study, we design an AV background environment with vivid and familiar working objects, such as servers, a CPU, memory, and computer networks, for network engineers to increase the realism of their virtual interaction with the network dashboard data. An AV-enhanced visualization dashboard may lead to an improved network-monitoring experience [46].

Studies that empirically examined the effectiveness of AV are few [47]. For instance, Gonzalez et al. [21] found that when AV is experienced simultaneously with real elements in the real world, it can increase user presence and reduce stress and simulation efficacy compared with a VR setting without real-world augmentation. Therefore, we infer that an AV environment with enhanced network-monitoring graphics (e.g., a CPU, servers, and network utilization) in a virtual dashboard environment can increase realism and users' sense of presence in the network environment. That is, such an AV environment will drive network professionals to become immersed in the resolution of their complex problems. In the literature, empirical studies that leveraged AV design with DD/UD virtualization designs in the context of complicated network-monitoring tasks, such as disaster recovery or emergency incident handling, are lacking. Therefore, we aim to explore whether user interactivity in an immersive AV environment will engross users in their critical and urgent problem-solving. Furthermore, we examine whether and how flow experience, enabled by AV and DD/UD designs, can lead to optimal user performance, which is important in a utilitarian environment.

2.3. Flow Theory

Flow refers to "the holistic experience that people feel when they act with total involvement" ([22], p. 36). Being "in flow," users will perceive an optimal experience that is deeply connected with their intrinsic motivation and enjoyment. Flow theory [22,48] is widely adopted by IS scholars to explain the fully immersive experience of users when they interact with computer-based systems [23–26]. The theory is also used



Fig. 1. Reality–Virtuality Continuum

to explain the cognitive absorption behavior of users in a state of deep involvement with software (e.g., [49]). Numerous studies have been conducted on the user flow experience (e.g., [50–52]) in the computing field owing to the value of flow theory in uncovering emotional and cognitive behaviors and designing optimal UX for modern digital products.

Dashboards allow users to not only monitor the status of the system but also facilitate problem-solving processes by drilling down into the data. Thus, the task types involved in the use of dashboards include monitoring and problem solving. Prior research supported the applicability of factors related to flow theory in task contexts involving exploration, learning, and problem solving [25,53]. Studies found that flow theory offers a reasonable lens to understand user perceptions and behaviors in the context of e-learning [54], online shopping [24], online searching [55], mobile banking [56], instant messaging [57], and problem solving [25].

The unified theory of acceptance and use of technology (UTAUT) model recognizes the importance of intrinsic motivations and suggests the inclusion of flow-related factors in the examination of user intention and behavior [58]. Prior studies on utilitarian systems also included flow-related factors as important determinants of behavioral intention and outcomes [24,59,60]. In addition, the human–computer interaction literature advocated the inclusion of design principles to enhance UX, regardless of the system type. UX design principles focus on providing an enjoyable and engaging experience and have broad applicability. For example, Bolen et al. [61] emphasized that given the importance of the flow perspective, researchers must consider related factors in the evaluation of work and leisure contexts.

Wu et al. [25] used tablets for problem solving and found that their navigation design and visual appearance can influence users' core flow experience, including their enjoyment, concentration, and control, and thus impact their perceived performance and problem-solving efficiency. Following this research stream [23–25], we focus on three core UXs that are the most relevant to the network-monitoring context: *perceived enjoyment, control*, and *concentration*. According to the environment psychology literature [62], perceived enjoyment and control are the result of user interaction with the external environment, and concentration reflects users' cognitive responses [24]. The three core flow experiences fit well in the network-monitoring problem-solving context.

Perceived enjoyment is defined as users' affective perception of pleasant and exciting interactions with a system [51]. Nah et al. [63] found that perceived enjoyment is valuable for capturing users' emotional flow component. Enjoyment has been validated for measuring intrinsic motivation to engage in an activity [51], such as web use [26,64]. In our study, the dashboard can provide certain hedonic elements; hence, we choose enjoyment for our evaluation to measure users' affective flow experience with the dashboard.

According to flow theory, concentration is a key indicator of being in a state of flow, given that users have a short attention span and limited time and cognitive resources for resolving their problems [25]. Users' concentration can positively affect their holistic experience of computer use and repeated use behavior [52]. Moreover, in the modern disruptive environment, concentrating on their current activity is difficult for users. Environmental and mental distractions can significantly limit users' ability to concentrate [25]. Therefore, understanding user concentration is critical in examining the effectiveness of dashboard designs that focus on critical problem solving for network outages and incidents.

Meanwhile, perceived control represents users' perception of their freedom and flexibility to act on their own [62]. This user perception is a core component of multiple theoretical frameworks, including self-efficacy theory [65] and the intention–behavior misalignment model [66]. In our study, perceived control is a key consideration in capturing user interaction with a dashboard as part of the flow experience.

3. Hypothesis Development

Overall, we examine the efficacy of UD and DD designs, along with AV, through the lens of flow theory. In Sections 3.1–3.3, we compare the differences between UD and DD designs in terms of perceived user enjoyment, concentration, and control, which are the three key dimensions of flow experience. In Section 3.4, we propose the nonlinear effect of AV. Fig. 2 presents our proposed research model. The design elements of a visualization dashboard differ in three aspects: ordering, messaging, and interactivity. A UD design features low ordering, low messaging, and high interactivity, whereas a DD design features high ordering, high messaging, and low interactivity. Such elements may influence user flow experience associated with perceived enjoyment, concentration, and control, which may lead to users' perceived performance.

3.1. Effect of UD (vs. DD) Design on Perceived Enjoyment

We expect users to perceive a higher level of enjoyment with a UD-based dashboard than with a DD-based design. An effective dashboard design requires designers to engage users by structuring the information clearly and effectively. Users' ability to select the order of the information in a UD-based dashboard will provide them with the flexibility to choose the information artifacts they deem relevant for task execution. When this approach is combined with focused messaging, the outcome will be a highly enjoyable experience for users [67]. By contrast, a high degree of messaging and ordering can cause attention diversion or the split attention effect. Specifically, a user will be inundated with the challenge of comprehending how to execute a task while trying to understand the messaging structure and the information being communicated through a lengthy message. That is, the display of a large amount of information on a fixed display (high ordering) will force users to spend more time than necessary in comprehending the information, which is not enjoyable. Users will perceive an experience as enjoyable when they can look at the data from different perspectives with only a few clicks. Data on demand offers an intuitive experience and makes the use of dashboards fun. When information is displayed in a user-friendly interface that allows users to get quick answers to questions, dashboards will become enjoyable to use.

Users seek pleasure in their interaction with systems and enjoy intriguing and exploratory experiences [49]. An important factor that can invoke feelings of pleasure in users in their interaction with a system is flexibility. When design choices limit flexibility, user interaction experience may be constrained. A UD design offers exploratory opportunities through an unordered approach and a messaging structure that encourages content seeking, if necessary, for the effective comprehension of a situation. A high level of interactivity has been found to influence perceived enjoyment and induce a high state of cognitive absorption [11,68,69]. Thus, allowing users to choose their own course of action and interaction patterns will engage them and induce an

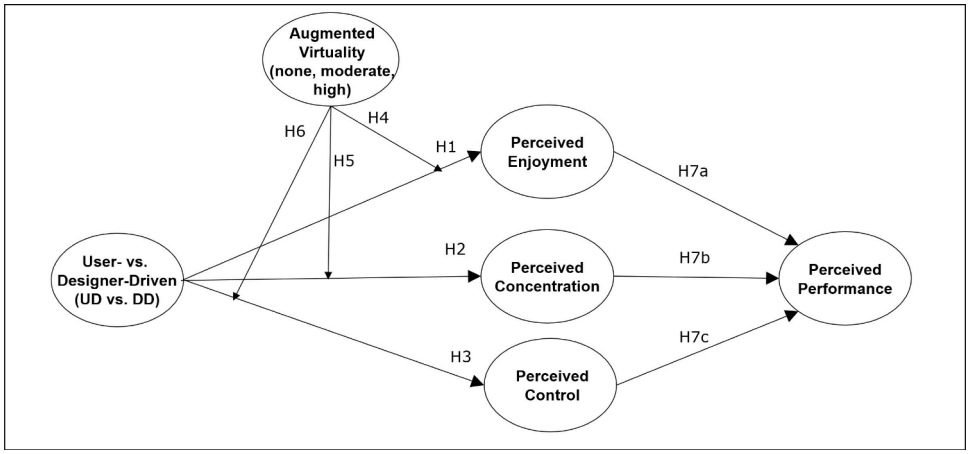


Fig. 2. Research Model

enjoyable experience. Therefore, we propose the following hypothesis:

Hypothesis 1. (H1): Users will experience higher perceived enjoyment in their use of a UD design dashboard than in their use of a DD design dashboard.

3.2. Effect of UD (vs. DD) Design on Perceived Concentration

A UD-based dashboard may lead to higher user concentration than a DD-based design. Concentration may facilitate complete immersion in an activity and represent an optimal state in which the mind focuses on a single task. Therefore, we postulate that the display of an excessive amount of content can create comprehension and concentration challenges, especially when urgency is required. The key concepts that can provide insights into the variable nature of concentration are interest and attention span. The preordering of content to include information deemed by the designer as appropriate and its delivery, with a low level of interactivity, can offer a preprogrammed experience. Interaction with a DD design dashboard may not trigger concentration because the preordered list will not give the user the flexibility to choose, which will diminish the benefits of relevant information artifacts that match the user's specific needs. Thus, a preordered list may not provide an immersive experience compared with an exploratory investigation to identify relevant information. When a UD design offers users the ability to discover unique paths that fit certain task purposes, it will likely capture their attention to access critical information and make timely decisions to resolve critical outage events. Information search behavior on a network-monitoring dashboard is similar to a general web search experience. UX can be improved through a design that offers users the ability to interact with information, based on their needs. Thus, design elements that provide options for how to interact with information and focused messaging will likely induce an immersive experience for users. Accordingly, we propose the following hypothesis:

Hypothesis 2. (H2): Users will experience higher concentration in using a UD design dashboard than in using a DD design dashboard.

3.3. Effect of UD (vs. DD) Design on Perceived Control

Perceived control captures a state in which a user feels free to act in ways they deem appropriate. In the website context, allowing users to select the order of the product information to browse freely will lead to a high level of control [35]. An individual's sense of control is related to their ability to exert influence [70]. Thus, inherent control conjures two entities (users and objects) and captures how they interact. In the context of this study, we refer to user interaction with a dashboard. A DD design is based on the premise that the designer explicitly understands the users' needs and builds the visual user interface on the basis of that understanding. Thus, a DD design structure may be aligned with user-perceived requirements but can come across as being too rigid, thereby limiting users' control over their interaction. By contrast, a UD design dashboard offers flexibility and can empower users to seek information through their flexible interaction with a system [71,72]. Thus, a UD design allows users to choose their own course of action and interaction patterns, which may result in a high level of control. Hence, we propose the following hypothesis:

Hypothesis 3. (H3): Users will experience more control in using a UD design dashboard than in using a DD design dashboard.

3.4. Effects of Three Levels of AV

In this study, we explore how AV works with UD and DD designs to engage users to efficiently resolve urgent network problems. We expect the effect of a UD (vs. DD) design on the three flow constructs to follow an inverted U-shaped curve as the AV level increases. AV is an amalgamation that can facilitate engagement in a virtual environment through the inclusion of real-world objects [73]. The integration of AV with different designs can alter UX. AV is expected to induce immersive UX [43], including realism, in a virtual environment compared with a traditional dashboard design. However, the role of AV is likely nonlinear.

Prior studies did not consider the role of design parameters in their examination of UX in AV versus non-AV environments. For example, how will visualization based on DD design principles and infused with AV impact users' flow experience? The impact may differ when AV is combined with a DD design versus a UD design. However, previous studies did not provide insights into such interactions and focused mostly on examining different user outcomes; Thus, the process through which the differences are manifested remains unclear [74]. The key features of a UD design are low messaging, low ordering, and high interactivity. The infusion of visual cues through AV will enhance messaging comprehension and augment interactivity through an experience infused with vivid and familiar objects [73]. Thus, a UD design will integrate well with AV to drive the flow experience.

An immersive AV environment will likely arouse high user interest in interacting with the data on the dashboard, and the higher the interest in the content, the better the comprehension and exploration [73]. Therefore, we expect the UX to be highly enjoyable [49]. Previous studies on 3D virtual worlds with Avatar and Second Life virtual environments, which are augmented with real-time communication and user interaction with 3D simulated characters, found positive and negative impacts on users' information-processing capability and attention span. For example, Nah et al. [75] compared the effects of 2D and 3D virtual worlds on brand equity in a 3D virtual hospital setting and found that a

3D virtual world environment can positively influence brand equity by fostering user flow experience. Meanwhile, Watson et al. [76] argued that virtual experience does not directly offer an interactive experience but a rich sensory one. The authors proposed that virtual augmentation can evoke a strong hedonic response when integrated with a system that provides high interactivity. Thus, in our study, the integration of a UD design with AV will likely make an experience highly enjoyable.

However, noting that flow experience lies within the boundaries of anxiety and boredom is important. Thus, a balanced approach to the use of AV may be necessary. Excessive reliance on AV may lead to distractions and thus result in high levels of user anxiety [75], whereas the use of few AV objects may not facilitate immersion and thus result in boredom. Thus, the role of AV in influencing the relationship between dashboard design elements and flow constructs could be nonlinear. A low AV level may not trigger an immersive experience, whereas a high AV level may distract users from completing their tasks. Overall, the nonlinear effect can be best described as the integration of a moderate level of AV, which can amplify the effect of the dashboard design compared with high or low levels of AV. Thus, a moderate level of AV will likely lead to higher perceived enjoyment than a non-AV design and help amplify the effects of a UD (vs. DD) design. Thus, we propose the following hypothesis:

Hypothesis 4. **(H4)**: With a moderate level of AV, the effect of a UD (vs. DD) design on perceived enjoyment will be more prominent than when the AV level is low or high.

Data visualization can show important facts about the relationship between imaginary graphics and real-world data [77]. The integration of AV with related content may create multiple channels (text and pictures; [78]) through which users can process the stimuli to foster an engaging experience [11]. As previously argued, a UD design can trigger high concentration through design features that encourage discovery. Such features will allow users to create their own paths to solve problems or make decisions. When design features are aided by a moderate level of AV, they may induce high levels of concentration in users. However, users' information-processing capability has limitations in a multisensory virtual environment, which can inhibit their attention. Dashboard designs with excessive graphics for connecting actual situations can distort critical messages. Yekta [79] argued that an inappropriate amount of multimodal input in a video learning task could cause cognitive overload that can hinder users' attentional focus, leading to decreased user comprehension. Nah et al. [75] found that a highly interactive and rich medium can result in attentional conflicts. Thus, the truth lies in the tradeoff to balance the design of multisensory modalities (e.g., visual and auditory) to increase available working memory resources. Therefore, we propose the following hypothesis:

Hypothesis 5. **(H5)**: With a moderate level of AV, the effect of a UD (vs. DD) design on user concentration will be more prominent than when the level of AV is low or high.

A UD design can provide users with a high sense of control by providing features that users can manipulate to complete a task. AV can provide a familiar environment that can enhance the sense of control when combined with new objects compared with an unfamiliar landscape. By expanding the processing channels and reducing the newness of a task through the addition of familiar objects, AV will likely enhance users' perception of control over their interaction with the dashboard [43]. UD features and AV will likely enhance a user's perceived control, and their impact on such a perception will likely be strong when they are combined in a singular UX with a dashboard. The key to understanding this linkage is the concept of temporal contiguity, which involves the introduction of familiar objects, along with new stimuli that can result in associative learning, to maintain the overall mental representation needed for effective comprehension [80]. Thus, users may exhibit a high level of control when they are introduced to multiple stimuli simultaneously (i.e., familiar UD features plus AV).

Owing to the rich and dynamic effects afforded by a high level of AV, when AV is high in a network-monitoring dashboard, its superior effect may be attenuated. Prior research showed that news playing in the background can impede users from completing complicated tasks on a website [81], and the richness and variety of attentional (e.g., audio, visual, and video) cues available in a 3D environment may inhibit users' attention. Thus, a moderate level of AV will work best, based on the premise that a balance must be maintained across the continuum of boredom and anxiety. Hence, we propose the following hypothesis:

Hypothesis 6. **(H6)**: With a moderate level of AV, the effect of a UD (vs. DD) design on perceived control will be more prominent than when the level of AV is low or high.

3.5. Effects of Flow on Perceived Performance

The effects of the three flow constructs on reuse intention are well established and justified in the literature [24], as well as the effects of the flow constructs on product understanding [23] and perceived performance [25]. When users enjoy interacting with a system, they will likely become involved in exploring its system features, which will result in a positive outcome. By contrast, a low level of enjoyment will turn users away, because they will not be intrinsically motivated to utilize the system. In addition, when users concentrate on certain tasks, they will filter out less relevant thoughts, which can improve their performance. Finally, having a high level of perceived control will boost users' confidence to effectively solve network-related problems, which will lead to improved performance. Thus, we propose the following hypotheses:

Hypothesis 7. **(H7)**: A high level of perceived enjoyment (**H7a**), perceived concentration (**H7b**), and perceived control (**H7c**) in using a visualization dashboard will lead to improved performance in diagnosing network issues.

4. Methodology

We designed a between-subjects 2 visualization approach (UD design vs. DD design) x 3 AV level (none, moderate, and high) factorial experiment for a network-monitoring dashboard interface to test the hypotheses. We followed a stepwise approach in our assessment. At the initial stage, we engaged a small sample of users in a pilot test for the UD and DD designs to refine our dashboard design parameters before adding the AV component. This approach enabled us to refine the study parameters and design the research for a large-scale empirical evaluation.

4.1. Manipulation of UD and DD Designs

The major network status parameters displayed on six dashboards were (i) CPU utilization, (ii) disk utilization, and (iii) network bandwidth availability. The information was centered on three different servers for a small-scale network monitoring environment. The states of the servers monitored on the dashboards were (i) critical, (ii) warning, and (iii) OK.

The design criteria we used to develop the UD-based network-monitoring dashboard were as follows: (i) a low degree of ordering, (ii) a low degree of messaging, and (iii) a high degree of interactivity. We achieved a low degree of ordering on the front end by providing a random sequence for the arrangement of the server details. We followed a sequence that offered users the ability to select/view the elements in a UD order. We reduced the messaging content to a low level and implemented a high degree of interactivity by providing a drill-down view and chart-type change options in the dashboard design (Appendix A). The drill-down view can help end users conduct an in-depth analysis of the device alert mechanism and support trend charts. The chart-type change option also contributed to the high-interactivity factor, which allowed end users to analyze the trend in the patterns of the individual servers in the form of a bar graph (Table 2).

Table 2
Summary of Specific Design Elements across Dashboards

Item	Messaging	Ordering	Interactivity
UD	(i) Status message, (ii) source IP address, (iii) device category, (iv) severity, (v) last check timestamp updated	(i) No predefined order for navigation, (ii) average disk, average CPU, and average network utilization statistics arranged in random sequence, which will guide toward specific alert details	(i) Drill-down option and (ii) trend chart-type change option
DD	(i) Status message; (ii) source IP address; (iii) device category; (iv) severity; (v) last check timestamp updated; (vi) detailed troubleshooting steps; (vii) impact, urgency, and outage action description; (viii) detailed device and alert description; (ix) outage action described in detail; (x) last timestamp check detailed information	(i) Predefined order for navigation, (ii) server needing attention will be displayed on left-most end, (iii) display sequence follows a descending pattern based on the priority of incident	(i) Performance trend chart for the last 24 hours for CPU, disk, and network utilization

Note: The content presented in the DD and UD designs is information equivalent. Although the DD design had more written messages than the UD design, we ensured that the additional messages were covered in other places in the UD dashboard, such as within the drill down.

The design criteria for the development of the DD-based dashboard were as follows: (i) linear ordering, (ii) a high degree of messaging, and (iii) a low degree of interactivity. The linear ordering pattern followed a predefined ordering sequence. The server needing attention was displayed on the left-most end, and the display sequence was based on the priority of the incident. An incident can be defined as any unexpected service interruption that can affect the normal state of the network status. We achieved a high degree of messaging by providing a large number of messages on the tactical overview table at the front end. We did not use a drill-down view and a chart-type change option in the DD design to ensure a low degree of interactivity (Table 2).

4.2. AV Manipulation

We manipulated three levels of AV (none, moderate, and high). The dashboard users should be able to relate the dashboard to the real world with the help of animation, static images, and background effects. For the dashboards with a moderate level of AV, we added static images representing the CPU, disk, and network availability to the dashboard background, from top to bottom. For the dashboards with a high level of AV, we rotated the same images in the dashboard background one at a time. Then, we integrated the AV design criteria with the UD and DD designs.

4.3. Main Experiment

Before the main experiment, we conducted a pilot study with 25 college students and 24 employees from network operations teams of a Fortune 500 data storage company to assess the psychometric properties of the constructs and ensure that the AV and dashboard design manipulations worked as intended. Appendix B lists the measurements for the manipulation checks and dependent variables. We adopted a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree) for the questions.

For the main experiment, we provided incentives such as course credits (1% of the final score of the course) to the study participants recruited from a large public university in the United States. We required the participants to first fill out a questionnaire to record their demographic data and the study's control variables. Then, we randomly assigned the participants to six experimental groups. Next, we instructed the participants to answer the questions related to the manipulation check and flow constructs.

The study sample consisted of 178 individuals recruited. The power analysis of a between-subjects design determined that 107 subjects (18 subjects in each group) would assure a sufficient statistical power of 0.80 to detect a medium effect size ($f = 0.30$; [82]). Among the participants, 44% were female, and the rest were male. The average age of the participants was 23.75 years. The participants perceived themselves to have excellent general computer skills and the ability to browse the web (5.88/7.00). We found no significant differences across the treatment conditions regarding the factors (i.e., gender, age, and skills). The results suggested that our random assignment of the participants to different experimental conditions was successful.

4.4. Data Analysis

First, we examined whether the manipulation of the UD versus DD design was successful[1]. We found that the participants perceived higher levels of ordering (mean difference = 1.3, $p < 0.001$) and messaging (mean difference = 1.4, $p < 0.001$) and a lower level of interactivity (mean difference = 1.8, $p < 0.01$) in the DD design than in the UD design. The results are consistent with those of the pilot study.

Second, the participants in the moderate AV level condition perceived a higher level of AV than those in the no AV condition (mean difference = 1.13, $p < 0.01$). That is, they noticed the images of the CPU, hard disk, and computer networks in the background of the dashboard. In addition, the participants in the high AV level condition perceived a higher level of AV than those in the moderate AV level condition (mean difference = 2.23, $p < 0.01$). That is, the participants noticed the rotated images of the CPU, hard disk, and computer networks in the background of the dashboard. We also found that the participant's perception of the (rotated) background images was correlated with their AV perception at $p < 0.05$, which lent support to our intended AV manipulation.

Table 3 shows the descriptive statistics of the three flow constructs. We examined the reliability and validity of the constructs. The reliability indicators measured by Cronbach's α (Table 3) were all above the cited minimum value of 0.7 [83]. To assess the discriminant validity, Fornell and Larcker [84] argued that the average variance extracted (AVE) should be larger than the variance shared between a construct and the other constructs in the model (i.e., the squared correlation between two constructs). Table 3 shows that all the construct pairs met this requirement. We further verified the discriminant validity by analyzing whether the loadings of the items on their targeted construct were higher than those on the other constructs in the model [85]. The exploratory factor analysis with SmartPLS demonstrated that each variable item loaded highly on its latent variable but less on the other variables (Table 4), which indicated adequate discriminant validity.

The independent sample t-test revealed that compared with the DD design, the UD design led neither to a higher level of perceived enjoyment (mean difference = 0.17, $p > 0.05$) nor to a higher level of perceived concentration (mean difference = 0.31, $p > 0.05$), thereby failing to support H1 and H2. However, the UD (vs. DD) design led to a

[1] The manipulation of the DD and UD conditions may result in unintended differences beyond those in ordering, messaging, and interactivity. We took special care to minimize the unintended differences and ensured that the two conditions were information equivalent. Although the DD design had more written messages than the UD design, we ensured that the additional messages were covered in other places of the dashboard, such as within the drill-down function. We also found no significant difference in the conditions ($p > 0.05$) regarding the effort spent and clarity of instruction presented on the dashboard.

Table 3
Descriptive Statistics and Correlations

Variable	Mean	Standard Deviation	Cronbach's α	1	2	3
1. Enjoyment	2.55	1.46	0.91	0.79*		
2. Concentration	3.15	1.44	0.84	0.71	**0.76**	
3. Control	3.22	1.59	0.83	0.60	0.58	**0.86**

* Note: Diagonal elements are square roots of AVE; off-diagonal elements are interconstruct correlations

Table 4
Factor Loadings and Cross-loadings

Construct	Items	Enjoyment	Concentration	Control	Performance
Enjoyment	Enjoyment1	**0.950**	0.668	0.569	0.495
	Enjoyment2	**0.935**	0.716	0.614	0.484
	Enjoyment3	**0.925**	0.732	0.527	0.500
	Enjoyment4	**0.942**	0.705	0.583	0.502
Concentration	Concentration 1	0.733	**0.875**	0.578	0.505
	Concentration 2	0.464	**0.785**	0.364	0.420
	Concentration 3	0.691	**0.870**	0.496	0.417
Control	Control 1	0.553	0.484	**0.913**	0.420
	Control 2	0.577	0.573	**0.938**	0.493
Performance	Performance1	0.411	0.433	0.414	**0.750**
	Performance2	0.490	0.498	0.455	**0.876**
	Performance3	0.369	0.347	0.321	**0.805**

significantly high level of perceived control (mean difference = 0.85, $p < 0.01$), thereby supporting H3.

To test H4–H6, we performed two sets of MANOVA. The first 2 visualization approach (DD design vs. UD design) x 2 AV level (none vs. moderate) MANOVA on the three flow constructs, with the control variables of age, gender, and general computer skills, indicated that the three interaction effects on perceived enjoyment, concentration, and control were significant at $p < 0.05$. As depicted in Figs. 3–5, with no AV, the difference between the DD and UD designs was minimal; however, the difference between the two designs was large when a moderate level of AV was integrated.

The second 2 visualization approach (DD design vs. UD design) x 2 AV level (moderate vs. high) MANOVA on the three flow constructs, with the same control variables, indicated that the interaction effects between the visualization approaches (DD design vs. UD design) and AV levels (moderate vs. high) on perceived enjoyment and concentration were significant at $p < 0.05$, and that on perceived control was marginally significant ($p = 0.07$). As depicted in Figs. 3–5, the significantly large difference between the DD and UD designs, with a moderate level of AV, became small with a high level of AV. That is, a high (vs. moderate) AV level reduced the superior effect of the UD design over the DD design. Overall, the two sets of MANOVA supported H4–H6. Regarding the effect of the flow constructs on perceived performance, the SmartPLS results revealed that perceived performance was significantly influenced by perceived enjoyment ($\beta = 0.19$, $p < 0.05$), perceived concentration ($\beta = 0.25$, $p < 0.01$), and perceived control ($\beta = 0.23$, $p < 0.01$), thereby supporting H7a–H7c. The three flow constructs jointly explained 35% of the variance in perceived performance. Table 6 summarizes the hypothesis testing results.

4.5. Supplementary Analysis of Objective Performance

In addition to measuring perceived performance, we asked the participants to answer four questions (Appendix B: Measurements) to measure objective performance. For each question, we coded the correct responses as 1 and the wrong ones as 0. Thus, the value of objective performance ranged from 0 to 4. The mean and standard deviation values of each group are reported in Table 5. The Pearson correlation showed a significant correlation between objective performance and perceived control ($\beta = 0.21$, $p < 0.01$), perceived concentration ($\beta = 0.22$, $p < 0.01$), perceived enjoyment ($\beta = 0.25$, $p < 0.001$), and perceived performance ($\beta = 0.31$, $p < 0.001$). In addition, the SmartPLS results indicated that objective performance was significantly affected by perceived enjoyment ($\beta = 0.21$, $p < 0.05$) but not by perceived control ($p > 0.05$) or concentration ($p > 0.05$). We believe that the strong correlation between perceived enjoyment and objective

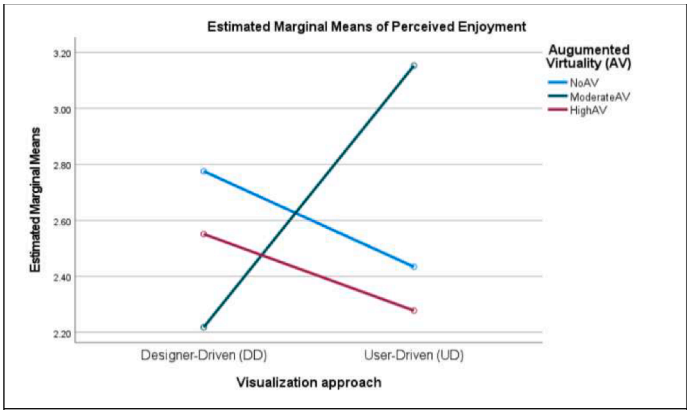


Fig. 3. Hypotheses Testing Results for H4
Support of H4 ($p<0.05$)

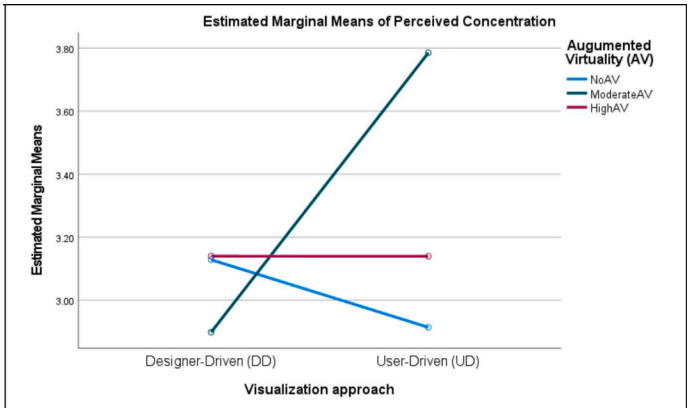


Fig. 4. Hypotheses Testing Results for H5
Support of H5 (p<0.05)

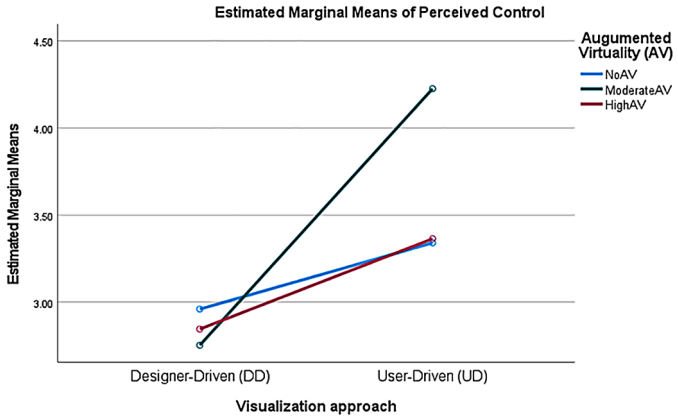


Fig. 5. Hypotheses Testing Results for H6
Support of H6 (p<=0.07)

performance dominated the effects of the other two flow constructs on objective performance. In other words, all three flow constructs are important, but perceived control and concentration are relatively less important in the presence of perceived enjoyment.

4.6. Post-hoc Analysis

We explored the possible correlations between the granular dashboard design elements and user flow experience. First, ordering, interactivity, and messaging were weakly correlated with the p-value of each pair of correlation coefficients higher than 0.50, thereby indicating that the three dimensions were distinct. Second, interactivity and ordering, but not messaging, had a significant influence on the three flow constructs. Specifically, interactivity and ordering had a stronger effect on perceived control ($p < 0.001$) than on enjoyment ($p < 0.01$) and on concentration ($p < 0.01$), which was evidenced by their level of significance.

The effects of the three flow constructs on reuse intention are well established and justified in the literature [24]. Therefore, we excluded them from our formal hypotheses for the sake of brevity. To confirm whether reuse intention was associated with the three flow constructs, we performed a correlation test and found that reuse intention was significantly correlated with perceived enjoyment ($r = 0.63, p < 0.001$), concentration ($r = 0.61$, $p < 0.01$), and control ($r = 0.55$, $p < 0.001$). Unsurprisingly, the results align with those of prior studies on the positive effect of flow on a series of outcomes, including product diagnosticity [23] and task performance [25].

5. Discussion and Conclusions

5.1. Discussion

The results indicate that a UD dashboard design can offer an improved solution in the network-monitoring context when combined with a moderate level of AV. Enhanced interactivity and low ordering are the key characteristics of a UD design that can offer improved flow experience in monitoring a network and reduce the complexity of understanding the visualization. The high interactivity of a UD design can provide users with the flexibility to choose a personal navigation path to understand the task on the dashboard. In addition, the results support the utilization of AV at a moderate level. As noted previously, a high level of AV may be perceived by users as a distraction, whereas a low level may not trigger the necessary immersion level to elicit attention. Thus, selecting the appropriate level of AV is important, because it can amplify the positive impact of dashboard design elements on users' flow experience or adversely affect the outcome of the interaction between the user and the system.

Table 5
Means (Standard Deviations) of Dependent Variables in Each Group

Visualization	AV	Enjoyment	Control	Concentration	Perceived Performance	Objective Performance
Designer-driven (DD)	**Low**	2.73 (1.51)	2.91 (1.64)	3.08(1.46)	3.86(1.22)	3.10(0.72)
	Medium	2.20(1.32)	2.73(1.47)	2.87(1.50)	3.78(1.38)	2.63(1.03)
	High	2.51(1.44)	2.79(1.50)	3.09(1.48)	3.66(1.45)	2.91(0.89)
User-driven (UD)	**Low**	2.49(1.49)	3.40(1.56)	2.97(1.27)	3.88(1.41)	3.06(1.12)
	Medium	3.14(1.58)	4.21(1.49)	3.77(1.46)	4.09(1.62)	3.17 (0.86)
	High	2.34 (1.36)	3.43(1.50)	3.21(1.37)	3.78(1.40)	3.16(1.11)
Subtotal of AVConditions	**Low**	2.60 (1.49)	3.16 (1.61)	2.60 (1.49)	3.87 (1.31)	3.08(0.94)
	Medium	2.65(1.51)	3.44(1.64)	2.65(1.51)	3.93 (1.50)	2.89 (0.89)
	High	2.43(1.39)	3.08(1.52)	2.43(1.39)	3.72(1.42)	3.02(0.99)
Subtotal of Visualization Designs	**DD**	2.47(1.42)	2.81(1.52)	3.02(1.47)	3.76 (1.35)	2.88 (0.90)
	UD	2.64 (1.50)	3.66(1.54)	3.33(1.39)	3.91(1.47)	3.13 (1.03)

Table 6
Summary of Hypothesis Testing Results

Hypothesis	Significance (p value)	Supported
H1: UD vs. DD —> enjoyment	NS	No
H2: UD vs. DD —> concentration	NS	No
H3: UD vs. DD —> control	< 0.05	Yes
H4. With a moderate level of AV, the effect of a UD (vs. DD) design on perceived enjoyment will be more prominent than when the level of AV is low or high.	< 0.05 for moderate vs. none and high vs. moderate	Yes
H5. With a moderate level of AV, the effect of a UD (vs. DD) design on user concentration will be more prominent than when the level of AV is low or high.	< 0.05 for moderate vs. none and high vs. moderate	Yes
H6. With a moderate level of AV, the effect of a UD (vs. DD) design on user control will be more prominent than when the level of AV is low or high.	< 0.05 for moderate vs. none; 0.07 for high vs. moderate	Yes
H7a: Perceived enjoyment —> perceived performance	< 0.05	Yes
H7b: Perceived concentration —> perceived performance	< 0.01	Yes
H7c: Perceived control —> perceived performance	< 0.01	Yes

NS: No significance

Segel and Heer [7] argued that a UD approach is better than a DD approach in supporting tasks such as data diagnostics, pattern detection, and hypothesis generation. Accordingly, we believe that a UD approach would be more appropriate than a DD approach for network-monitoring administrators to diagnose network issues and develop solutions to resolve identified network problems. We find only the overall positive effect of the UD (vs. DD) design on perceived control. Given that the overall main effects on enjoyment and concentration are not significant, we analyze whether the level of AV plays a nonlinear role in influencing the relationship between the visualization approaches (UD design vs. DD design) and the three flow constructs. The findings show that the UD approach can influence the users' flow experience more than the DD approach only when the AV level is moderate. Thus, the nonsignificant main effect may be due to the combination of the sample with no AV and moderate and high levels of AV [86].

UD design features can enable users to interact with the dashboard based on their preferences. In a stressful context, such an interaction will likely invoke a strong perception of being in control (H3), rather than enjoyment or an immersive experience under any circumstances. Thus, features associated with a UD design put users in control of their interaction with the dashboard. However, enjoyable and immersive experiences require the augmentation of the basic UD design with a moderate level of AV. That is, a UD design alone will not provide significant differentiation from a DD design in terms of enjoyment and concentration but must be combined with a moderate level of AV to have a significant impact. AV elements will likely provide contextual cues that can offer highly enjoyable and immersive experiences when combined with UD design elements.

The attenuated positive effect of a UD (vs. DD) approach when the level of AV is high can result from a situation in which individuals are presented with material that exceeds their working memory, which can deteriorate the learning effect [87]. Accordingly, a high level of AV, with multiple rotating background hardware images, may place a burden on users to process excessive incoming information and thus reduce the positive effect of the UD (vs. DD) design. When the AV level is moderate, users' cognitive load will remain at a reasonable level during the diagnostic process. Thus, they can take advantage of a UD approach to interact with the dashboard and diagnose network issues. Specifically, with a moderate level of (vs. no) AV, users can relate the presented information on the dashboard to the real world, which will result in increased perceived realism and reduced stress [21].

5.2. Theoretical Contributions

Our study makes multiple theoretical contributions to the literature. First, our study contributes to the visual design literature. Network outages and problems require immediate attention from network personnel. In an emergency with time pressure, whether a DD design or a UD design will work better is unclear. Specifically, our study examines the differences between a UD design and a DD design in terms of enjoyment, control, and concentration and thus offers a complete and fine-grained understanding of the two visualization styles. As such, our study advances the literature by empirically examining the differences between the two approaches in the network visualization domain.

Second, we extend flow theory to the context of network-monitoring dashboard designs and examine how flow experience can be improved through different combinations of visualization approaches (UD and DD designs) and AV. Flow theory is typically employed in the e-commerce context [24,35] but rarely used in the dashboard design domain. Thus, understanding how design elements can improve user engagement in the context of systems that aid mission-critical network monitoring is limited. We propose two design concepts (the visualization approach and AV) as antecedents of flow constructs and provide insights into user interaction with the designs. Our assessment opens up further discussion on the need to obtain deep insights into the relationship between design elements and users' cognitive (e.g., concentration) and affective (e.g., enjoyment) processes. In terms of the direct effects, we find that a design that can offer an aggregated overview, with high flexibility, can invoke a high sense of control in users.

Third, we reveal that the downstream effect of the three flow constructs is not trivial in the utilitarian context. Perceived enjoyment, concentration, and control can significantly affect users' performance in diagnosing network problems correctly. Contrary to the conventional wisdom that flow is important only in the usage context of hedonic systems, we find that concentration, control, and enjoyment are also conducive to users' performance in using a utilitarian system. That is, distraction, loss of control, and high stress can negatively affect users' state of mind when learning about network monitoring, identifying the cause of a network problem, and formulating the necessary steps to resolve the network problem.

Fourth, our study contributes to the emerging AV literature. Recently, the AIS community called for IS-related research on AV [21]. Our results reveal that the relationship between the visualization approach (DD and UD designs) and flow constructs may not be linear but may depend on the appropriate AV level. Thus, finding the correct balance with the cues embedded in the virtual environment is important. Determining whether the inclusion of additional cues to aid users will be beneficial or overwhelming is challenging. The inclusion of additional cues in the virtual environment can increase the probability of users' attention to be compromised, which can lead to less-than-optimal experiences. Our study validates this argument because the rotating images embedded in the dashboard weakened the positive impact of the design elements on the participants' flow experience. From a theoretical standpoint, recognizing that a rich medium will not consistently lead to a highly immersive experience for the users is important.

AV will likely influence a user's cognitive processing, leading to a flow experience, owing to its interesting and novel approach of adding real physical objects to a virtual environment [88]. Moreover, the augmentation of the real world and virtual world can create unique personal experiences for users and facilitate their interaction in the virtual environment. This can draw users' attention and lead them to become deeply engrossed in an activity, that is, in the flow situation, which was considered by Wu et al. [25] to be the most optimal situation for the best user performance. We find empirical evidence that AV plays different roles in leveraging DD/UD designs to effectively facilitate user engagement to obtain critical outcomes. Thus, our study opens up a new research stream to advance our understanding of how AV can be

leveraged to improve user performance in complicated and urgent situations.

In summary, our study makes significant contributions to the HCI literature through important findings and implications. We provide design guidelines to the IT industry for using AV to leverage DD/UD designs for critical and time-sensitive problem-solving tasks for optimal user performance. Our work also lays the foundation for new research to advance reality-based HCI and IS research, which is critical for future technology design and development to support utilitarian tasks and enhance problem-solving quality in time-sensitive environments. Our study is generalizable because it is applicable to different problem-solving design contexts on digital platforms. The AV design is a trade-off to contextualize the study context for optimal user performance, facilitated by user flow experience. This finding is important to reality-enabled technology designs for reshaping the future of work, driven by artificial intelligence (AI).

5.3. Practical Contributions

From a practical standpoint, our study offers insights to designers to explore elements that may expand users' processing channels. As previously suggested, users can open up multiple channels when the stimuli are delivered in different formats (audio and picture). This perspective can offer insights into why AV plays an important role in enhancing UX with dashboard design elements. Our study results show that with a moderate level of AV, the positive effect of a UD (vs. DD) design can be fully employed to engage users in the complicated real-time network-monitoring context.

However, this expansion is limited. Users may experience attention overload if exposed to overwhelming stimuli. Familiarity with objects can help mitigate this effect; hence, stimuli that will appeal to multiple channels must be included in a purposeful manner. Many environments have shifted to digital platforms; thus, understanding how to effectively use AV concepts is important for designers. Our results refute the notion that more is better. Designers must be selective and mindful in implementing AV to avoid overwhelming users. A key difference between the AV design elements (moderate vs. high AV level) in the experiment is the moving objects. In AV, motion in elements is not an ideal design, especially in systems that aid mission-critical infrastructure monitoring.

5.4. Limitations and Future Research

Our study has certain limitations. First, we focus only on UD and DD designs. Expanding the design elements to include other design concepts (e.g., mixed UD and DD designs) and assessing their role in network-monitoring dashboard designs and performance would be interesting. Second, though our study provides solid evidence based on users' perceptions, we failed to capture in-depth psychological data. For example, measuring users' attention, cognitive load, and emotions during their interaction with a network-monitoring dashboard using eye-tracking, facial reader, galvanic skin response, and electroencephalography sensors can help in designing an in-depth biometric study approach. Such techniques can be used to capture users' gaze, brain activities, and emotional responses to provide further evidence and insights into which dashboard design would be optimal for user performance. The assessment can help outline the boundary conditions to customize UX for improved job performance, which is directly related to dealing with complicated network- monitoring outages and maintaining regular network infrastructure operations. In addition, examining the role of different design approaches and AV from multiple theoretical perspectives is important. For example, our current approach is based on flow theory, but future studies can explore whether flow theory or other theories can offer a better theoretical lens through which to explore phenomena and relationships.

We manipulated AV through the addition of image items of network devices, a CPU, and the memory as the dashboard background. From this lightweight AV manipulation, we found that the high level of AV overwhelmed the users and led to suboptimal flow experience and user performance. We expect an alternative AV manipulation, showing the dynamic real-time and frequently updated information from multiple network-monitoring devices, to further increase the users' cognitive load, which might negatively affect their performance in stressful situations. Nevertheless, we call for future research to explore this possibility.

Our study is one of the first attempts to understand how AV can be integrated with visualization approaches to improve network-monitoring dashboard designs. Therefore, we expect advanced technological development to be achieved through further research, especially on how AI technologies can be used to capture real-time network dynamics on network infrastructures and how AI-based technologies (e.g., machine learning algorithms) can be used effectively to capture and provide early warnings to network administrators to be proactive in protecting the safety of the overall infrastructure. Meanwhile, an AV-based dashboard system can serve as a useful training tool for network engineers and data analysts to work together seamlessly to optimize IT operations to generate competitive business advantages.

CRediT authorship contribution statement

Khawaja A. Saeed: Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing. **Dezhi Wu:** Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing. **David Jingjun Xu:** Conceptualization, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing.

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Appendix A. Screenshot of UD and DD Conditions without AV (Figs. A1–A2) and DD Conditions with a Moderate AV Level (Fig. A3)

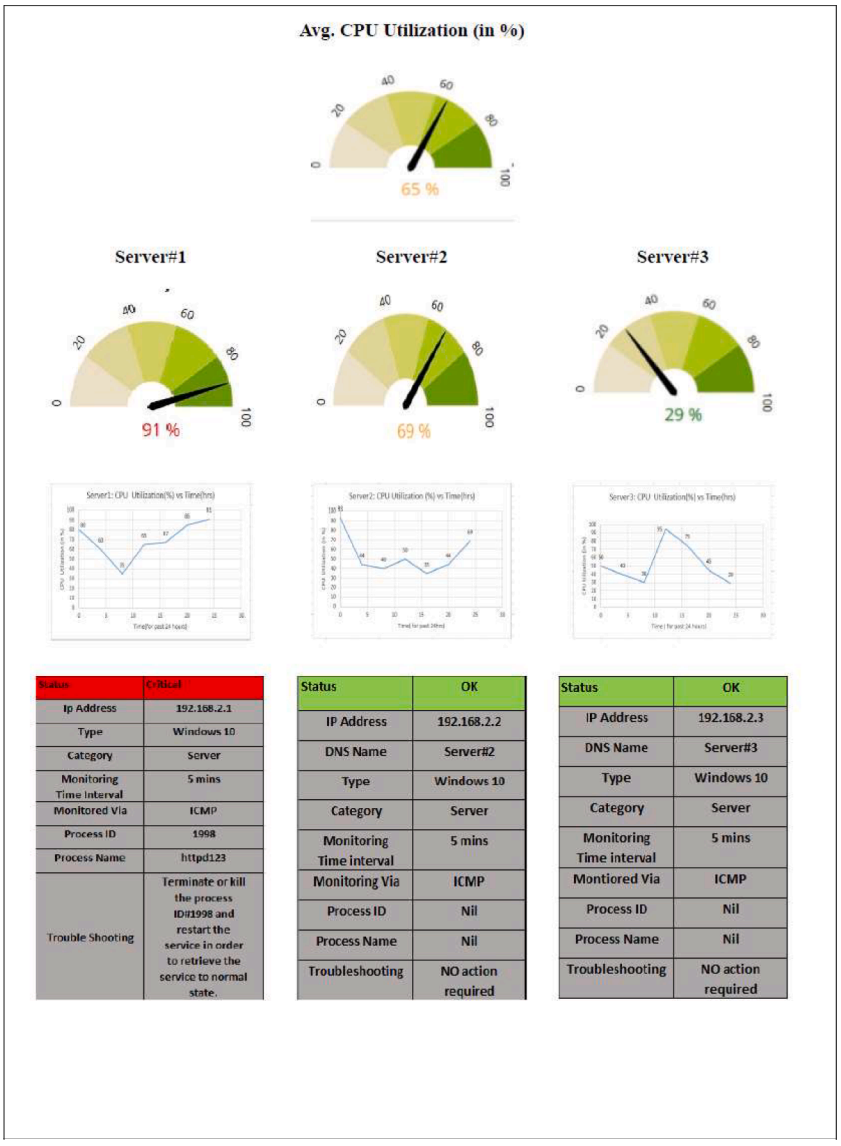


Network Monitoring Dashboard

Tactical Overview

Message	Source IP address	Device Category	Severity	Last Check - Time Stamp Updated
Problem: CPU Utilization is high on Server#1 due to the Process#id1pid123, Process#id21598. Troubleshooting: The process which consumes more CPU has to be identified and killed in order to resume the service back to normal state. Impact: Affects all the hosts connected to this server. Urgency: The first response to be sent out within 15 minutes. OutageAction: Outage Notification has to be sent if the issue is not resolved within 4mins.	192.168.2.3 IP Address: An IP address is an identifier for a computer or device on a TCP/IP network. Networks using the TCP/IP protocol route messages based on the IP address of the destination. To view your IP address you can use the Ipconfig (IPCONFIG) command line tool. Ipconfig displays all current TCP/IP network configuration values	CPU Central processing unit (CPU) device is the hardware within a computer that carries out the instructions of a computer program by performing the basic arithmetical, logical, control and input/output operations of the system. CPU Alert: When a system is oversubscribed, however, and one or two sites experience a typical heavy loads, resulting in too many CPU processing requests, other Web sites housed on the same server could suffer a loss in	Critical Critical alert happens when a device does not respond to a ping. Critical: This state occurs whenever the utilization percentage exceeds 80%. Steps to be taken: (a) Create IT service remedy ticket (b) Input the alert details (c) Open the bridge call (d) Follow the troubleshooting guide (e) Resolve the ticket within SLA target	8 April 2017 07:31:11 PM The server status will be checked and updated in every 5 minutes. The above time stamp was the last updated status with respect to the CPU utilization of server#1.
Problem: Disk Space Utilization on server#1 needs attention. Diskid#1001 may lead to an outage. Troubleshooting: Data backup recovery process has to be done in order to avoid 'Critical' outage in future. Impact: Restricts from saving more data into the concerned disk. Urgency: The issue has to be fixed first. Outage Action: No need to send any outage communication until it reaches critical state.	192.168.2.1 IP Address: An IP address is an identifer for a computer or device on a TCP/IP network. Networks using the TCP/IP protocol route messages based on the IP address of the destination. To view your IP address you can use the ipconfig (IPCONFIG) command line tool. Ipconfig displays all current TCP/IP network configuration values	Disk Disk drive stores and provides relatively quick access to large amounts of data on an electromagnetically charged surface or set of surfaces. Disk space usage is a function of three variables: (a)Number and size of client workspaces (b)Size of server database (c) Size of server's archive of all versioned files.	Warning Warning alert happens when the response time exceeds a defined time limit. Warning: The utilization is between 70% and 80%. Steps to be taken: (a) Alert the team about the Warning status (b) Take Immediate back up recovery action (c) Reinstate the service back to normal.	8 April 2017 07:31:11 PM The server status will be checked and updated in every 5 minutes. The above time stamp was the last updated status with respect to the disk utilization of server#1.

Fig. A1. DD design without AV (content is presented vertically on a single webpage)

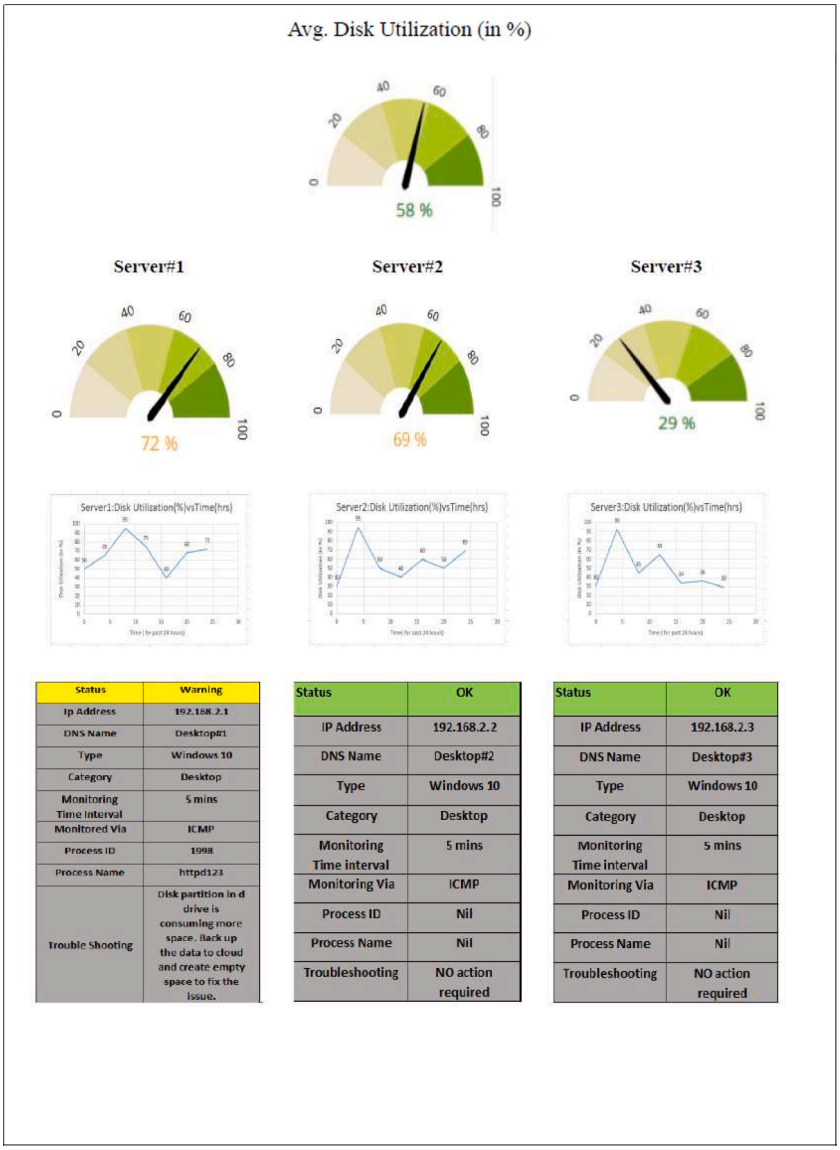


Status	Critical
Ip Address	192.168.2.1
Type	Windows 10
Category	Server
Monitoring Time Interval	5 mins
Monitored Via	ICMP
Process ID	1998
Process Name	httpd123
Trouble Shooting	Terminate or kill the process ID#1998 and restart the service in order to retrieve the service to normal state.

Status	OK
IP Address	192.168.2.2
DNS Name	Server#2
Type	Windows 10
Category	Server
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Status	OK
IP Address	192.168.2.3
DNS Name	Server#3
Type	Windows 10
Category	Server
Monitoring Time interval	5 mins
Montiored Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Fig. A1. (*continued*).

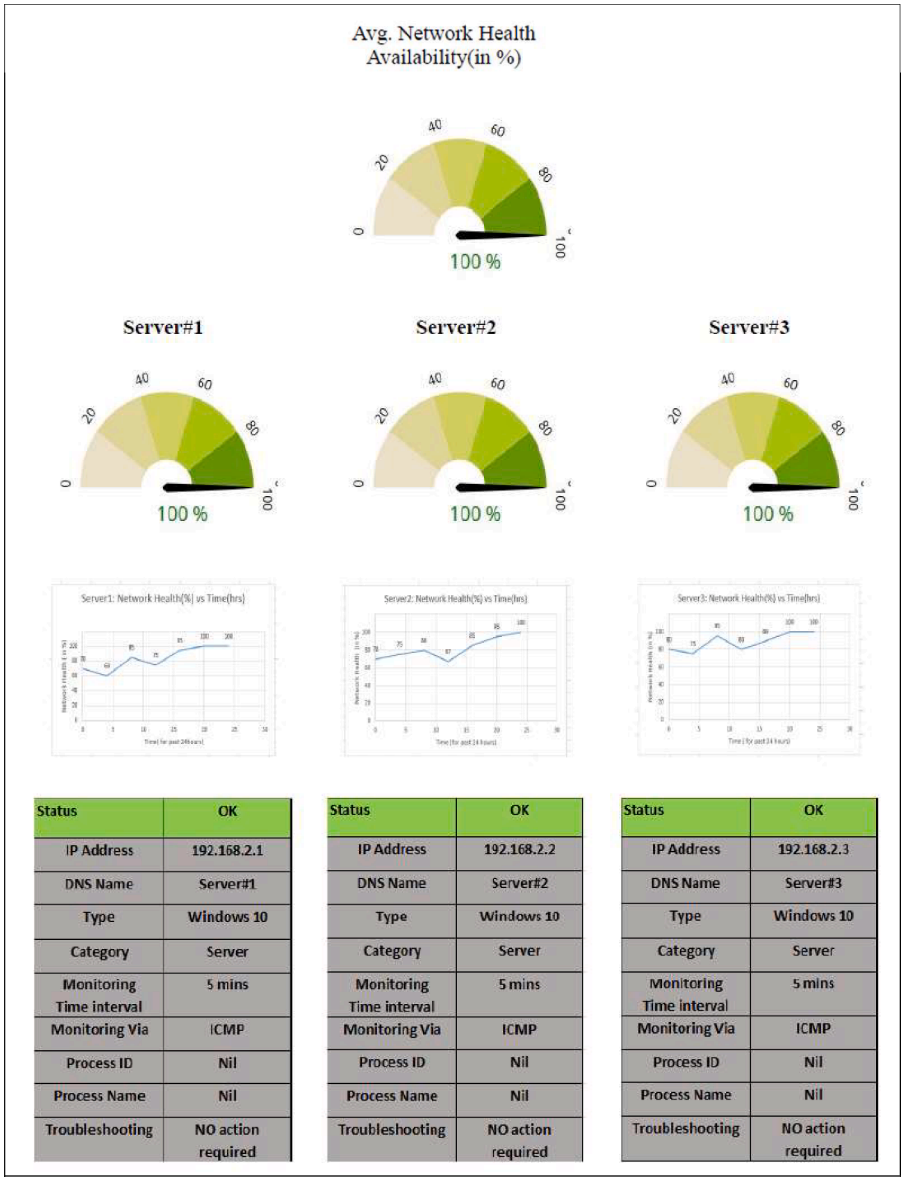


Status	Warning
Ip Address	192.168.2.1
DNS Name	Desktop#1
Type	Windows 10
Category	Desktop
Monitoring Time Interval	5 mins
Monitored Via	ICMP
Process ID	1998
Process Name	httpd123
Trouble Shooting	Disk partition in d drive is consuming more space. Back up the data to cloud and create empty space to fix the issue.

Status	OK
IP Address	192.168.2.2
DNS Name	Desktop#2
Type	Windows 10
Category	Desktop
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Status	OK
IP Address	192.168.2.3
DNS Name	Desktop#3
Type	Windows 10
Category	Desktop
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Fig. A1. (*continued*).



Status	OK
IP Address	192.168.2.1
DNS Name	Server#1
Type	Windows 10
Category	Server
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Status	OK
IP Address	192.168.2.2
DNS Name	Server#2
Type	Windows 10
Category	Server
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Status	OK
IP Address	192.168.2.3
DNS Name	Server#3
Type	Windows 10
Category	Server
Monitoring Time interval	5 mins
Monitoring Via	ICMP
Process ID	Nil
Process Name	Nil
Troubleshooting	NO action required

Fig. A1. (*continued*).

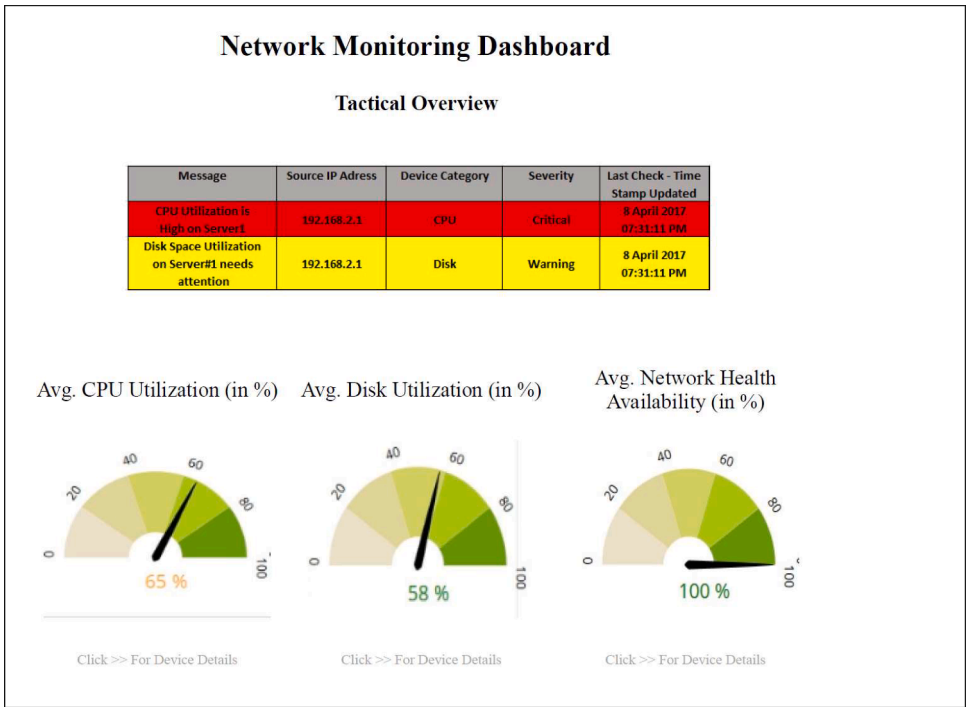


Fig. A2. **a.** UD design without AV (homepage with three click buttons [for device details] leading to three subpages shown in Figs. 2b–2d); **b.** UD design without AV (subpage 1 showing server CPU utilization); **c.** UD design without AV (subpage 2 showing server disk utilization); **d.** UD design without AV (subpage 3 showing network health)

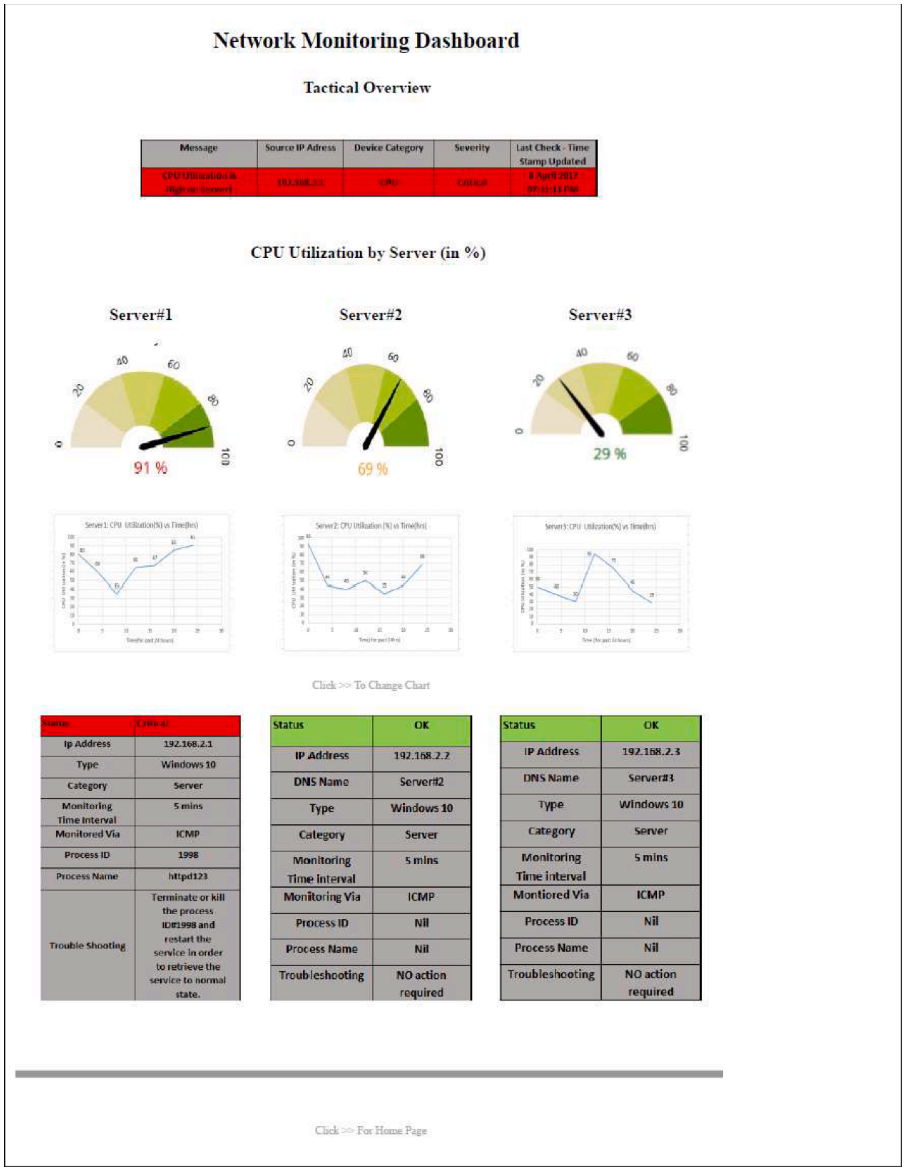


Fig. A2. (*continued*).

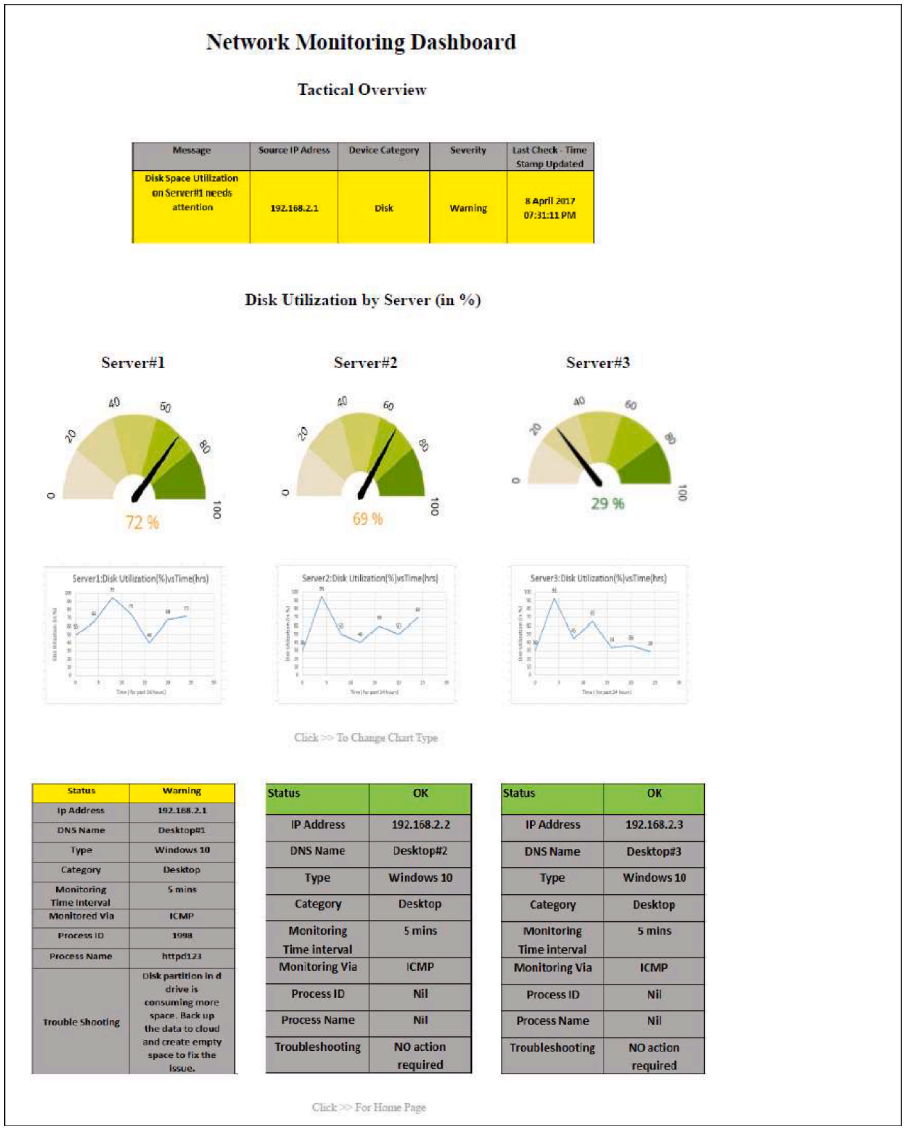


Fig. A2. (*continued*).

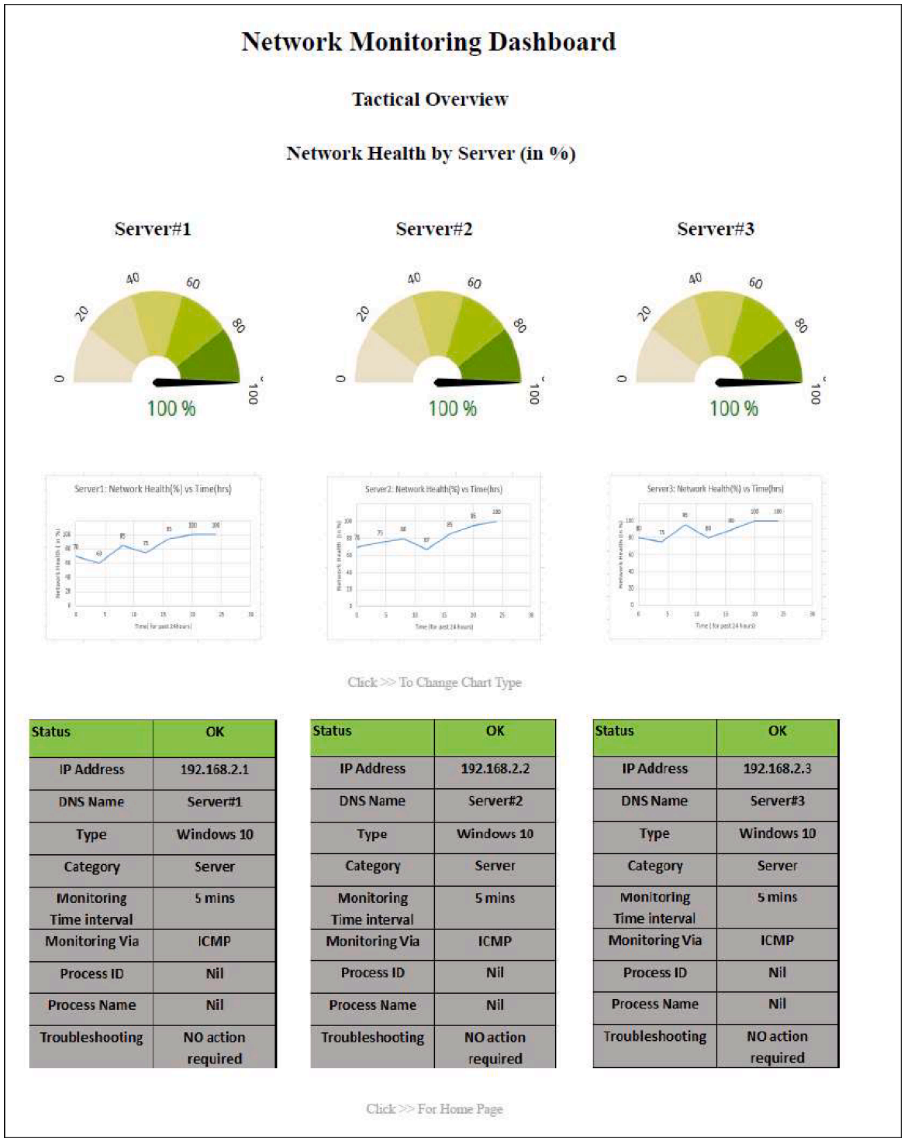


Fig. A2. (*continued*).



Fig. A3. **a**. UD design with a moderate level of AV (using static images as background; homepage with three click buttons [for device details] leading to three subpages shown in Figs. 3b–3d); **b**. UD design with a moderate level of AV (subpage 1 showing server CPU utilization); **c**. UD design with a moderate level of AV

(Subpage 2 showing server disk utilization); **d**. UD design with a moderate level of AV (subpage 3 showing network health)
Note 1: Figs. 2–3 are a UD design, because 1) the number of messages on the tactical overview table of the dashboard is few, and 2) the interactivity is high, because three buttons at the bottom of the page are available to click to show additional content; thus, users can choose the viewing order of the content.
Note 2: Fig. 3 shows the UD design with a moderate level of AV, because the webpage background includes images of the memory, a CPU and, network devices. The DD design with a moderate level of AV shows the images of a CPU, the memory, and the servers vertically on the same webpage.
Note 3: When the AV level is high, the images of the CPU, memory, and servers are rotated on the same webpage space.

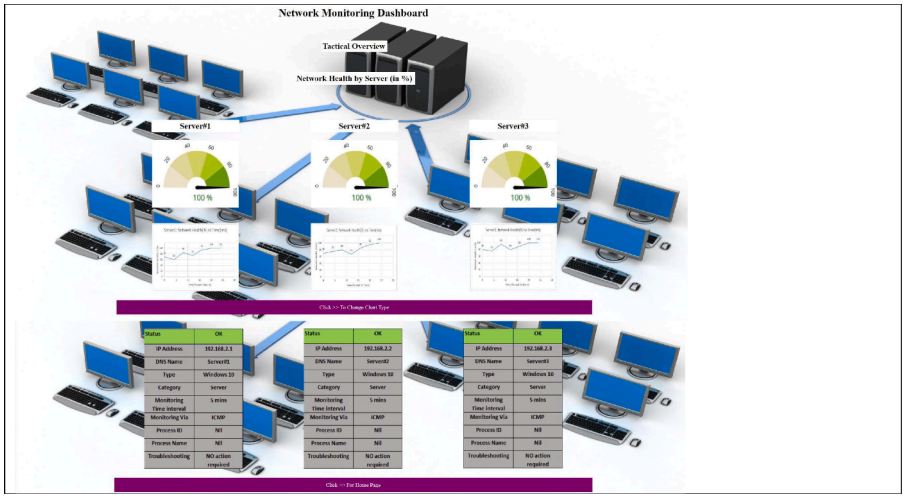


Fig. A3. (*continued*).

Appendix B. Measurements

Construct	Sample Measurement Items	Sources
Perceived ordering	• The dashboard presented information in a predefined order.	Segel and Heer [7]
Perceived messaging	• The number of written messages on the dashboard was large.	
Perceived interactivity	• I was able to interact with the dashboard.	Segel and Heer [7]; Jiang and Benbasat [11]
	• The level of interactivity of the dashboard was high.	
Manipulation check for AV	• The dashboard background had an image of the hard drive.	Self-developed
	• The dashboard background had an image of the CPU.	
	• The dashboard background had flashing images of the hard drive, the CPU, and interlinked computers.	
Perceived AV	• The dashboard had a real-world network-monitoring environment.	Albert et al. [43]
Enjoyment	• Using the dashboard provided me with considerable enjoyment.	Koufaris [24]
	• Using the dashboard for the network-monitoring task was exciting.	
	• Sometimes, I lose track of time when I am using the dashboard.	
	• Using the dashboard for the network-monitoring task was interesting.	
Concentration	• By using the dashboard, I was able to concentrate fully on the network-monitoring task.	
	• While using the dashboard, I was absorbed in what I was doing.	
	• By using the dashboard, I was able to deeply engage in the network-monitoring task.	
Control	• While using the dashboard, I felt in control.	
	• The dashboard allowed me to be in control.	
	• I felt that I had no control over my interaction with the dashboard (reverse coded and dropped).	
Perceived performance	• The dashboard assisted me in learning the cause of the network problem.	Turpin and Scholer [89]
	• I diagnosed the network problem accurately.	
	• I am confident in my assessment of the network problem.	
Objective performance	• When was the alert section "last updated" on the dashboard? (Options: Nov. 10, 2017, 9:31:11 pm; April 8, 2017, 7:31:11 pm; April 12, 2017, 6:31:11 pm; Nov. 14, 2017, 5:31:11 pm)	Self-developed
	• What is your overall observation of the network status? (Options: OK, warning, critical, critical and warning)	
	• Which device is in the "critical" state (red alert)? (Options: CPU, router, server, disk)	
	• Which device is in the "attention" state (yellow alert)? (Options: CPU, router, disk, switch)	

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